

Sets and Set Operations

Homework Presentation

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Mathematics for Computing
2nd August 2026



Set Builder Form

Agenda

1. Set Builder Form
2. Types of Sets
3. Set Operations
4. Hasse Diagrams
5. Generalized Inclusion–Exclusion Principle

Set Builder Form

Definition

A set-builder form describes a set by specifying the property that its elements satisfy.

$$A = \{x \mid x \text{ satisfies a given condition}\}$$

where

- x is an element.
- " $\mid$ " means "such that".
- The condition defines the members of the set.

Examples

1. Even numbers less than 20

Set Builder: $A = \{x \in \mathbb{N} \mid x < 20, x \text{ is even}\}$

Roster: $A = \{2, 4, 6, 8, 10, 12, 14, 16, 18\}$

2. Odd numbers less than 15

Set Builder: $B = \{x \in \mathbb{N} \mid x < 15, x \text{ is odd}\}$

Roster: $B = \{1, 3, 5, 7, 9, 11, 13\}$

3. Multiples of 5 up to 50

Set Builder: $C = \{x \in \mathbb{N} \mid x = 5n, 1 \leq n \leq 10\}$

Roster: $C = \{5, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$

4. Prime numbers less than 30

Set Builder: $D = \{x \in \mathbb{N} \mid x < 30, x \text{ is prime}\}$

Roster: $D = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$

5. Perfect squares less than 100

Set Builder: $E = \{x \in \mathbb{N} \mid x = n^2, 1 \leq n \leq 9\}$

Roster: $E = \{1, 4, 9, 16, 25, 36, 49, 64, 81\}$

Examples

6. Integers between -5 and 5
Set Builder: $F = \{x \in \mathbb{Z} \mid -5 \leq x \leq 5\}$
Roster: $F = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$
7. Positive multiples of 3
Set Builder: $G = \{x \in \mathbb{N} \mid x = 3n, n \in \mathbb{N}\}$
Roster: $G = \{3, 6, 9, 12, 15, \dots\}$
8. Numbers divisible by both 2 and 3
Set Builder: $I = \{x \in \mathbb{N} \mid x \text{ is divisible by 2 and 3}\}$
Roster: $I = \{6, 12, 18, 24, 30, \dots\}$
9. Alphabets that are vowels
Set Builder: $I = \{x \in \{a, b, \dots, z\} \mid x \text{ is a vowel}\}$
Roster: $I = \{a, e, i, o, u\}$
10. Multiples of 7 less than 50
Set Builder: $J = \{x \in \mathbb{N} \mid x = 7n, 1 \leq n \leq 7\}$
Roster: $J = \{7, 14, 21, 28, 35, 42, 49\}$



Types of Sets

Types of Sets

Topics Covered

- Cardinality of Set
- Finite Set
- Infinite Set
- Null Set
- Equal Set
- Equivalent Set
- Subset
- Superset
- Power Set
- Singleton Set
- Universal Set
- Countable Set

1. Cardinality of Set

Definition

The **cardinality** of a set is the number of distinct elements present in the set.

Mathematical Representation

$$|A| = n$$

Example

$$A = \{1, 2, 3, 4, 5\}, \quad |A| = 5$$

AI/ML Example

A training dataset

$$D = \{\text{Image}_1, \text{Image}_2, \dots, \text{Image}_{1000}\}$$

contains 1000 images. Hence,

$$|D| = 1000$$

2. Finite Set

Definition

A **finite set** is a set containing a fixed (limited) number of elements.

Mathematical Definition

$$A = \{a_1, a_2, \dots, a_n\}, \quad |A| = n, \quad n \in \mathbb{N}$$

Example

$$A = \{2, 4, 6, 8, 10\}, \quad |A| = 5$$

AI/ML Example

Set of ML algorithms:

$$M = \{\text{CNN, SVM, Decision Tree, Random Forest}\}$$

Hence,

$$|M| = 4$$

3. Infinite Set

Definition

An **infinite set** is a set that contains infinitely many elements.

Mathematical Definition

$$A = \{a_1, a_2, a_3, \dots\}, \quad |A| = \infty$$

Example

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

AI/ML Example

The set of all possible feature vectors is

$$F = \{x \mid x \in \mathbb{R}^n\}$$

Hence,

$$|F| = \infty$$

4. Null Set

Definition

A **null (empty) set** is a set containing no elements.

Mathematical Definition

$$A = \emptyset \quad \text{or} \quad A = \{\}, \quad |A| = 0$$

Example

$$A = \{x \in \mathbb{N} \mid x < 0\} = \emptyset$$

AI/ML Example

Let

$$C = \{\text{Images classified as both Cat and Car}\}$$

If no such image exists, then

$$C = \emptyset$$

5. Equal Set

Definition

Two sets are called **equal** if they contain exactly the same elements.

Mathematical Definition

$$A = B \iff (\forall x)(x \in A \iff x \in B)$$

Example

$$A = \{1, 2, 3\}, \quad B = \{3, 2, 1\}$$

Hence, $A = B$.

AI/ML Example

$$L_1 = \{\text{Cat}, \text{Dog}, \text{Bird}\}$$

$$L_2 = \{\text{Bird}, \text{Dog}, \text{Cat}\}$$

Hence, $L_1 = L_2$.

6. Equivalent Set

Definition

Two sets are **equivalent** if they have the same cardinality (same number of elements).

Mathematical Definition

$$A \sim B \iff |A| = |B|$$

Example

$$A = \{1, 2, 3\}, \quad B = \{a, b, c\}$$

Hence, $|A| = |B| = 3$ and $A \sim B$.

AI/ML Example

$$|D_1| = |D_2| = 1000$$

Hence, $D_1 \sim D_2$.

7. Subset

Definition

A set A is a **subset** of a set B if every element of A is also an element of B .

Mathematical Definition

$$A \subseteq B \iff (\forall x)(x \in A \Rightarrow x \in B)$$

Example

$$A = \{1, 2\}, \quad B = \{1, 2, 3, 4\}$$

Hence, $A \subseteq B$.

AI/ML Example

Let

$$A = \{\text{Cat Images}\}$$

$$B = \{\text{Animal Images}\}$$

Hence, $A \subseteq B$.

8. Superset

Definition

A set B is called a **superset** of a set A if every element of A is also an element of B .

Mathematical Definition

$$B \supseteq A \iff (\forall x)(x \in A \Rightarrow x \in B)$$

AI/ML Example

Let

$A = \{\text{Cat Images}\}$

$B = \{\text{Animal Images}\}$

Then,

$$B \supseteq A$$

9. Power Set

Definition

The **power set** of a set A is the set containing all possible subsets of A .

Mathematical Definition

$$P(A) = \{X \mid X \subseteq A\}$$

Cardinality

$$|P(A)| = 2^{|A|}$$

AI/ML Example

Let

$$F = \{\text{Age}, \text{Income}\}$$

Then,

$$P(F) = \{\emptyset, \{\text{Age}\}, \{\text{Income}\}, \{\text{Age}, \text{Income}\}\}$$

(All possible feature combinations used for feature selection.)

10. Singleton Set

Definition

A **singleton set** is a set containing exactly one element.

Mathematical Definition

$$A = \{x\} \quad \text{where} \quad |A| = 1$$

AI/ML Example

A classification model predicts only one class.

$$\text{Prediction} = \{\text{Cat}\}$$

Hence,

$$|\text{Prediction}| = 1$$

11. Universal Set

Definition

A **universal set** is the set containing all elements under consideration.

Mathematical Definition

$U = \{x \mid x \text{ belongs to the universal domain}\}$

If A is any set, then

$$A \subseteq U$$

AI/ML Example

Let

$U = \{\text{All Images in ImageNet}\}$

Every image dataset is a subset of U .

For example,

$D = \{\text{Cat Images}\}$

Hence,

$$D \subseteq U$$

12. Countable Set

Definition

A **countable set** is a set whose elements can be listed in a sequence or put into one-to-one correspondence with the natural numbers.

Mathematical Definition

$$A = \{a_1, a_2, a_3, \dots\}$$

Examples:

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

AI/ML Example

A training dataset can be indexed as

$$D = \{d_1, d_2, d_3, \dots, d_n\}$$

Hence, the dataset is countable.

Summary

Type	Symbol
Cardinality	$ A $
Finite	$ A < \infty$
Infinite	$ A = \infty$
Null	$\emptyset$
Equal	$A = B$
Equivalent	$ A = B $
Subset	$A \subseteq B$
Superset	$A \supseteq B$
Power Set	$P(A)$
Singleton	$ A = 1$
Universal	U
Countable	$\mathbb{N}, \mathbb{Z}$



Set Operations

Set Operations

Common Set Operations

- Union
- Intersection
- Set Difference
- Complement
- Cartesian Product
- Symmetric Difference

1. Union

Definition

The **union** of two sets contains all elements that belong to either set.

Mathematical Definition

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

AI/ML Example

Let

$$A = \{\text{Cat Images}\}$$

$$B = \{\text{Dog Images}\}$$

Then,

$$A \cup B = \{\text{Cat or Dog Images}\}$$

2. Intersection

Definition

The **intersection** of two sets contains only the elements common to both sets.

Mathematical Definition

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

AI/ML Example

Let

$A = \{\text{Cat Images}\}$

$B = \{\text{Black Images}\}$

Then,

$$A \cap B = \{\text{Black Cat Images}\}$$

3. Set Difference

Definition

The **set difference** of two sets contains the elements of one set that are not present in the other.

Mathematical Definition

$$A - B = \{x \mid x \in A \wedge x \notin B\}$$

AI/ML Example

Let

$T = \{\text{Training Images}\}$

$C = \{\text{Corrupted Images}\}$

Then,

$$T - C = \{\text{Valid Training Images}\}$$

4. Complement

Definition

The **complement** of a set A is the set of all elements in the universal set U that are not in A .

Mathematical Definition

$$A^c = U - A = \{x \mid x \in U \wedge x \notin A\}$$

AI/ML Example

Let

$$A = \{\text{Spam Emails}\}$$

Then,

$$A^c = \{\text{Legitimate Emails}\}$$

where A^c represents all non-spam emails in the dataset.

5. Cartesian Product

Definition

The **Cartesian product** of two sets A and B is the set of all ordered pairs formed by taking one element from A and one from B .

Mathematical Definition

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

AI/ML Example

Let

$$F = \{\text{Images}\}, \quad L = \{\text{Labels}\}$$

Then,

$$F \times L = \{(\text{Image}_1, \text{Cat}), (\text{Image}_2, \text{Dog}), \dots\}$$

Each ordered pair represents an image and its corresponding label used for training a machine learning model.

6. Symmetric Difference

Definition

The **symmetric difference** of two sets contains elements that belong to either set, but not to both.

Mathematical Definition

$$A \triangle B = (A - B) \cup (B - A)$$

AI/ML Example

Let

$F_A = \{\text{Features selected by Algorithm A}\}$

$F_B = \{\text{Features selected by Algorithm B}\}$

Then,

$$F_A \triangle F_B = \{\text{Features selected by only one algorithm}\}$$

Comparison of Set Operations

Operation	Symbol	Meaning
Union	$A \cup B$	All elements
Intersection	$A \cap B$	Common elements
Difference	$A - B$	Only in A
Complement	A^c	Not in A
Cartesian Product	$A \times B$	Ordered pairs
Symmetric Difference	$A \Delta B$	In one set only



Hasse Diagrams

Hasse Diagram

Definition

A Hasse Diagram is a graphical representation of a finite partially ordered set (Poset).

It represents the ordering relation without showing reflexive and transitive edges.

Applications

- Database Hierarchies
- Knowledge Representation
- Artificial Intelligence
- Ontology Design
- Feature Hierarchies in Machine Learning

Question 1: Hasse Diagram

Given

$$A = \{1, 2, 3, 4, 5\}, \quad aRb \iff a \leq b$$

Properties

- Reflexive
- Antisymmetric
- Transitive

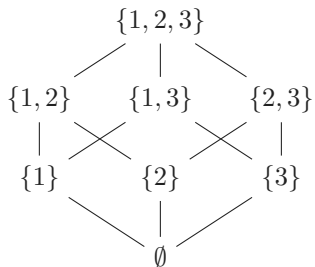
Hence, $(A, \leq)$ is a **Poset**.

5
|
4
|
3
|
2
|
1

Question 2: Hasse Diagram

Given

$$A = \{1, 2, 3\}, \quad S_i R S_j \iff S_i \subseteq S_j$$





Generalized Inclusion–Exclusion Principle

Generalized Inclusion–Exclusion Principle

Theorem

For finite sets

$$A_1, A_2, \dots, A_n,$$

the number of elements in their union is

$$\left| \bigcup_{i=1}^n A_i \right| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} \left| \bigcap_{i=1}^n A_i \right|.$$

Idea

Each element is counted exactly once by alternately adding and subtracting overlaps.

Proof of Generalized Inclusion–Exclusion Principle

Let $A_1, A_2, \dots, A_n$ be finite sets and suppose an element x belongs to exactly r of these sets.

Then,

- In $\sum |A_i|$, the element x is counted $\binom{r}{1}$ times.
- In $\sum |A_i \cap A_j|$, it is counted $-\binom{r}{2}$ times.
- In $\sum |A_i \cap A_j \cap A_k|$, it is counted $+\binom{r}{3}$ times.
- The process continues by alternately subtracting and adding higher-order intersections.
- Thus, the total count of x becomes

$$\binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \dots + (-1)^{r+1} \binom{r}{r}.$$

Proof of Generalized Inclusion–Exclusion Principle (Contd.)

Hence, the total number of times the element x is counted is

$$\binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \cdots + (-1)^{r+1} \binom{r}{r}.$$

Using the binomial identity,

$$\binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \cdots + (-1)^{r+1} \binom{r}{r} = 1.$$

Therefore, every element is counted exactly once, giving

$$\left| \bigcup_{i=1}^n A_i \right| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} \left| \bigcap_{i=1}^n A_i \right|.$$

Hence, the Generalized Inclusion–Exclusion Principle is proved.

Thank You

Department of Computer Science